

A REALIZATION OF G_2 FOR LITT PROBLEM NO. 13

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ABSTRACT. We construct a complex abelian threefold P and an integral symmetric ample divisor $X \subset P$ whose full convolution Tannaka group is the simple group of type G_2 . The construction starts from a cyclic triple cover $C \rightarrow E$ of an elliptic curve, equivalently from a smooth curve $C : w^6 = f_3(x)$. The Prym polarization has type $(1, 1, 3)$, and the theta section X has a unique simple elliptic singularity of type $\tilde{E}_7$. Its intersection complex $K = \mathrm{IC}_X$ satisfies

$$\chi(K) = 7, \quad K \text{ is a direct summand of } \mathrm{Alt}^2(K).$$

Here the second assertion is in the convolution Tannakian category; it is obtained from an exact universal norm-conic identity and a support calculation for pair convolution. The tautological representation is faithful, orthogonal, and irreducible on the identity component; its Hodge cocharacter has multiplicities $(2, 3, 2)$. These data identify the full group, including its component group, with G_2 in its standard seven-dimensional representation.

1. INTRODUCTION

Let A be a complex abelian variety. Convolution of constructible complexes on A induces a neutral Tannakian structure after quotienting by the negligible objects. Thus an integral subvariety $Z \subset A$ with nonzero intersection-cohomology Euler characteristic determines an algebraic group $G(\mathrm{IC}_Z)$ and a faithful representation of dimension $\chi(\mathrm{IC}_Z)$; see [13]. Litt Problem No. 13 asks whether the exceptional simple groups

$$G_2, \quad E_7, \quad E_8, \quad F_4$$

occur in this way [16].

This paper settles the G_2 case. Let

$$\mathcal{B} = \mathrm{Spec} \mathbf{C}[A, B, (4A^3 + 27B^2)^{-1}].$$

For $b = (A, B) \in \mathcal{B}(\mathbf{C})$, consider the smooth projective curves

$$E_b : y^2 = x^3 + Ax + B, \quad C_b : w^6 = x^3 + Ax + B,$$

and the cyclic triple covering

$$\pi_b : C_b \longrightarrow E_b, \quad (x, w) \longmapsto (x, w^3).$$

Write $O \in E_b$ for the point at infinity, put

$$\kappa_b = \pi_b^* \mathcal{O}_{E_b}(O), \quad P_b = \ker(\mathrm{Nm}_{\pi_b})^0,$$

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and define the cyclic theta section

$$X_b = \{M \in P_b : h^0(C_b, \kappa_b \otimes M) > 0\}.$$

Theorem 1.1 (Main theorem). *There is a nonempty Zariski-open subset $\mathcal{B}^\circ \subset \mathcal{B}$ such that, for every $b \in \mathcal{B}^\circ(\mathbf{C})$, the variety P_b is an abelian threefold, $X_b \subset P_b$ is an integral symmetric ample divisor, and, as an algebraic group together with its tautological module,*

$$(G(\mathrm{IC}_{X_b}), \omega(\mathrm{IC}_{X_b})) \simeq (G_2, V_7),$$

where V_7 is the standard faithful seven-dimensional representation. This is the full convolution Tannaka group, not only its identity component or its connected derived group.

The proof has three parts. First, the geometry of the norm fibre gives a minimal resolution $\rho : S \rightarrow X$ with

$$R\rho_* \mathbf{Q}_S[2] \simeq \mathrm{IC}_X \oplus \delta_0, \quad \chi(\mathrm{IC}_X) = 7,$$

and the generic-twist Hodge vector of IC_X is $(2, 3, 2)$. Second, a universal norm calculation shows that the generic pair-incidence curve

$$C_x = \{y \in X : -x - y \in X\}$$

has precisely two nonexceptional components, globally labelled over a dense open subset of X , and transposition exchanges them. A top-cohomology rank calculation and the decomposition theorem then give

$$\mathrm{IC}_X \subset \mathrm{Alt}^2(\mathrm{IC}_X).$$

Finally, the Hodge cocharacter and the exterior-square inclusion distinguish G_2 from the other connected irreducible orthogonal subgroups of O_7 . The same exterior-square inclusion excludes the only possible disconnected extension.

The universal norm identity is an exact calculation over $\mathbf{Q}(\lambda, e_2, e_3, u)$, not a numerical experiment. The supplementary verification performs rational-function arithmetic and independently checks that a cleared polynomial numerator is identically zero. Finite-field calculations are used only to certify that specified Zariski-open conditions are nonempty.

All varieties are over $\mathbf{C}$. Constructible complexes and Hodge modules retain their canonical $\mathbf{Q}$ -structures. Every Tannaka group and tautological module below means the corresponding object after extension of scalars from $\mathbf{Q}$ to $\mathbf{C}$; in particular, all algebraic groups are over $\mathbf{C}$.

2. TANNAKIAN CONVENTIONS

Let A be a complex abelian variety and let $m : A \times A \rightarrow A$ be addition. For constructible complexes we write

$$K * L = Rm_*(K \boxtimes L).$$

Perverse sheaves of Euler characteristic zero generate the negligible tensor ideal. The quotient of the semisimple perverse category by this ideal is rigid symmetric monoidal, and the tensor subcategory generated by one nonnegligible object is neutral Tannakian [13, 20].

For an integral subvariety $Z \subset A$ of dimension d , let

$$\mathrm{IC}_Z = j_{!*} \mathbf{Q}_{Z_{\mathrm{reg}}}[d]$$

be its perverse intersection complex, extended by zero to A . If $\chi(\mathrm{IC}_Z) > 0$, we denote by $G(\mathrm{IC}_Z)$ the complexification of the Tannaka group of the category generated by IC_Z , and by $\omega(\mathrm{IC}_Z)$ its complex tautological module. Then

$$\dim \omega(\mathrm{IC}_Z) = \chi(\mathrm{IC}_Z).$$

The action on the tautological module is faithful by construction.

The commutativity constraint contains the usual cohomological sign. Since the support in this paper has even dimension two, the alternating projector on $\mathrm{IC}_X * \mathrm{IC}_X$ is carried by the fibre functor to the ordinary exterior square of $\omega(\mathrm{IC}_X)$. We write $\mathrm{Alt}^2(\mathrm{IC}_X)$ for its image in the Tannakian quotient.

3. THE CYCLIC THETA SECTION

We suppress the parameter b until Section 7. Thus

$$E : y^2 = f_3(x), \quad C : w^6 = f_3(x), \quad y = w^3,$$

where f_3 is a separable cubic, and $\pi : C \rightarrow E$ is the resulting cyclic triple cover.

3.1. The cover, the Prym, and the polarization. The three branch points of π are the nonzero points of $E[2]$. If $\eta = \mathcal{O}_E(O)$, then the cyclic covering algebra is

$$\pi_* \mathcal{O}_C = \mathcal{O}_E \oplus \eta^{-1} \oplus \eta^{-2}.$$

Riemann–Hurwitz gives $g(C) = 4$, while the cyclic-cover canonical bundle formula gives

$$K_C = \pi^*(K_E \otimes \eta^2) = \pi^* \eta^2 = \kappa^2, \quad \kappa = \pi^* \eta.$$

Projection formula yields

$$h^0(C, \kappa) = h^0(E, \eta) + h^0(E, \mathcal{O}_E) + h^0(E, \eta^{-1}) = 2.$$

The pullback $\pi^* : J(E) \rightarrow J(C)$ is injective. Indeed, if $\alpha \in \mathrm{Pic}^0(E)$ and $\pi^* \alpha \simeq \mathcal{O}_C$, then

$$1 = h^0(C, \pi^* \alpha) = h^0(E, \alpha) + h^0(E, \alpha \eta^{-1}) + h^0(E, \alpha \eta^{-2}),$$

and the last two terms vanish; hence $\alpha = \mathcal{O}_E$. Duality now shows that $\ker(\mathrm{Nm}_\pi)$ is connected. We therefore write

$$P = \ker(\mathrm{Nm}_\pi).$$

It has dimension three.

Use the theta characteristic κ to centre the theta divisor on $J(C) = \mathrm{Pic}^0(C)$:

$$\Theta_\kappa = \{M \in J(C) : h^0(C, \kappa \otimes M) > 0\}.$$

Let $L = \mathcal{O}_{J(C)}(\Theta_\kappa)|_P$. The standard polarization formula for a degree-three cyclic cover ramified at three points gives

$$\mathrm{type}(L) = (1, 1, 3), \quad L^3 = 3! \chi(L) = 18;$$

see [14, Section 4] and [2, Corollary 12.1.4 and Lemma 12.3.1]. Scheme-theoretically,

$$X = \Theta_\kappa \cap P = \{M \in P : h^0(C, \kappa \otimes M) > 0\}.$$

Thus $X \in |L|$ and is ample. Serre duality and $\kappa^2 = K_C$ give, for $M \in P$,

$$h^0(C, \kappa \otimes M) = h^0(C, \kappa \otimes M^{-1}),$$

and therefore

$$X = -X$$

with centre the origin. The uniqueness of this centre will follow from (3.2).

3.2. The norm fibre and the resolution. Let

$$n : C^{(3)} \longrightarrow \text{Pic}^3(E), \quad D \longmapsto \mathcal{O}_E(\pi_* D),$$

and put

$$N = n^{-1}(\eta^3), \quad a : N \longrightarrow X, \quad D \longmapsto \mathcal{O}_C(D) \otimes \kappa^{-1}.$$

The pencil $\Gamma = |\kappa| \simeq \mathbf{P}^1$ will be shown below to be the only positive-dimensional fibre of a . We record the precise resolution because its topology determines both the Euler characteristic and the Hodge vector.

Let r_1, r_2, r_3 be the ramification points and $D_0 = r_1 + r_2 + r_3 \in \Gamma$. In local coordinates at the three branch points, the norm map is analytically

$$(x_1, x_2, x_3) \longmapsto x_1^3 + x_2^3 + x_3^3.$$

Thus (N, D_0) is the affine cone over a smooth plane cubic, a simple elliptic singularity of type $\tilde{E}_6$ in the standard classification [15, 18].

We next check that there are no other singularities. Away from the ramification points the differential of the norm map is nonzero. At a ramification point, the local covering is $z \mapsto z^3$. For a divisor with two points colliding there, in elementary symmetric coordinates one has

$$z_1^3 + z_2^3 = e_1^3 - 3e_1e_2,$$

which has no linear term; with three points colliding, however,

$$z_1^3 + z_2^3 + z_3^3 = e_1^3 - 3e_1e_2 + 3e_3$$

has a nonzero linear term. Hence the only possible critical divisors in N are D_0 and $2r_i + r_j$ with $i \neq j$. The latter would require $2b_i + b_j \sim 3O$. Since the b_i are the nonzero two-torsion points, this would give $b_j = O$, a contradiction. Thus D_0 is the unique singular point of N .

Blowing up its vertex now gives a smooth surface T with exceptional elliptic curve E_0 satisfying $E_0^2 = -3$. The strict transform $\tilde{\Gamma}$ meets E_0 transversely once.

The symmetric-product canonical formula [17] gives $K_{C^{(3)}} = a_C^* \Theta_C$, and adjunction on the norm fibre gives $K_N \cdot \Gamma = 0$. Since the singularity has multiplicity three,

$$K_T = \sigma^* K_N - E_0.$$

Adjunction on $\tilde{\Gamma}$ then yields $\tilde{\Gamma}^2 = -1$. Contracting $\tilde{\Gamma}$ produces a smooth surface S ; the image E_{exc} of E_0 satisfies

$$E_{\text{exc}}^2 = -2.$$

Finally, $S \rightarrow X$ contracts E_{exc} and is the minimal resolution. The standard classification of simple elliptic singularities therefore identifies the singularity of X at the origin as type $\tilde{E}_7$ [15, 18].

For completeness, the Euler characteristic of N can be computed by the finite map $N \rightarrow |\eta^3| \simeq \mathbf{P}^2$. Stratifying a line section of the plane cubic by its multiplicity

pattern gives the following table.

base divisor type	$e(\text{stratum})$	# fibre	contribution
three distinct points, no branch point	14	27	378
three distinct points, one branch point	-12	9	-108
the branch divisor	1	1	1
$2q + r$, $q, r \notin B$	-24	18	-432
$2b_i + r$	3	3	9
$2q + b_i$	12	6	72
$3q$ at a flex	9	10	90
			10

The tangent incidences in this table are distinct: a line through a branch point b_i and tangent at q satisfies $2q = -b_i$, and multiplication by two on E is étale of degree four. No nonzero two-torsion point is a flex. Together with the critical-point calculation above, this proves

$$e(N) = 10, \quad e(T) = 9, \quad e(S) = 8.$$

The canonical map of C is birational: on the affine chart it is

$$(x, w) \mapsto [w^2 : w : 1 : x].$$

Hence C is not hyperelliptic. Its canonical quadric is the rank-three quadric determined by the pencil $|\kappa|$, which has only one ruling. Consequently κ is the unique g_3^1 on C . Riemann–Kempf [11] shows that the Abel map is an isomorphism over the $h^0 = 1$ locus. The corresponding points of N are smooth by the critical-point calculation above, so their images in X are smooth. The pencil $|\kappa|$ is the only positive-dimensional Abel fibre. It follows that

$$\text{Sing}(X) = \{0\}.$$

The smooth dense open $X \setminus \{0\}$ shows that the Cartier divisor X is generically reduced; since a Cartier divisor on the smooth variety P is Cohen–Macaulay, X is reduced. An ample divisor on an abelian variety is connected. If X were reducible, two components would meet in a curve, which would lie in $\text{Sing}(X)$. Thus X is integral.

The map $\rho : S \rightarrow X$ is semismall, with the single relevant exceptional fibre E_{exc} . Its one-dimensional top intersection form is the nondegenerate matrix (-2) . The semismall decomposition theorem [4] gives

$$R\rho_* \mathbf{Q}_S[2] \simeq \text{IC}_X \oplus \delta_0.$$

Taking Euler characteristics in (3.2) and using (3.2), we obtain

$$\chi(\text{IC}_X) = 7.$$

The unique singular point also gives

$$\text{Stab}_P(X) = \{0\},$$

because a translation preserving X must preserve its singular locus. In particular, the origin is the unique centre of the symmetry (3.1): two distinct centres would compose to a nontrivial translation stabilizing X .

We will also use dominance of the conormal Gauss map. Here is a local-index calculation of its degree. The normal form in the simple elliptic classification shows

that a general hyperplane section of the $\tilde{E}_7$ germ is of type A_3 . Its sectional Milnor number is therefore 3, and the local Euler obstruction is $1 - 3 = -2$ [15, 18, 3]. On the other hand, (3.2) gives $\chi((\mathrm{IC}_X)_0) = -1$. The Dubson–Kashiwara local index formula [10] therefore gives

$$CC(\mathrm{IC}_X) = \Lambda_X + \Lambda_0.$$

The Franecki–Kapranov index theorem [6], together with (3.2), yields

$$\deg(\Lambda_X \rightarrow \mathbf{P}(T_0^*P)) = 6.$$

3.3. The generic-twist Hodge vector. We next determine the Hodge cocharacter needed in the group-theoretic step. The canonical divisor of N is the pullback of L , so $K_N^2 = L^3 = 18$. The blowup of the $\tilde{E}_6$ vertex gives

$$K_T^2 = K_N^2 + E_0^2 = 15,$$

and $T \rightarrow S$ is the blowup of a smooth point. Hence

$$K_S^2 = 16, \quad e(S) = 8, \quad \chi(\mathcal{O}_S) = 2,$$

where the last equality is Noether’s formula.

Intersection-cohomology weak Lefschetz for the ample divisor $X \subset P$ [1], together with (3.2), identifies

$$H^1(P, \mathbf{Q}) \xrightarrow{\sim} IH^1(X, \mathbf{Q}) \xrightarrow{\sim} H^1(S, \mathbf{Q}).$$

Thus $q(S) = 3$ and $\mathrm{Pic}^0(P) \rightarrow \mathrm{Pic}^0(S)$ is an isogeny. The morphism $S \rightarrow P$ has a two-dimensional image, so S has maximal Albanese dimension.

Choose a general unitary rank-one local system $\mathcal{L}_\alpha$ on P , with associated line bundle $\alpha \in \mathrm{Pic}^0(P)$. Coherent generic vanishing [8] gives

$$H^i(S, K_S \otimes \alpha|_S) = 0 \quad (i > 0).$$

Since $\chi(\mathcal{O}_S) = 2$, Riemann–Roch and Serre duality, applied also to α^{-1} , give

$$h^0(S, K_S \otimes \alpha|_S) = 2, \quad h^2(S, \alpha|_S) = 2.$$

Let $j : S \rightarrow P$ be the proper morphism used below. The complex $Rj_*\mathbf{Q}_S[2]$ is perverse: first $R\rho_*\mathbf{Q}_S[2]$ is perverse because ρ is semismall, and then extension along $X \hookrightarrow P$ is perverse t -exact. Generic vanishing for perverse sheaves on abelian varieties [13, 20] therefore gives

$$H^k(S, j^*\mathcal{L}_\alpha) = 0 \quad (k \neq 2).$$

For a unitary local system, the Hodge decomposition is the twisted Dolbeault decomposition

$$H^{p,q}(S, j^*\mathcal{L}_\alpha) = H^q(S, \Omega_S^p \otimes \alpha|_S), \quad p + q = 2.$$

The total dimension is $e(S) = 8$; the two extreme dimensions are both 2, so the Hodge vector of $Rj_*\mathbf{Q}_S^H[2]$ is $(2, 4, 2)$. The Hodge-module refinement of (3.2), extended to P , is

$$Rj_*\mathbf{Q}_S^H[2] \simeq \mathrm{IC}_X^H \oplus \mathbf{Q}_0^H(-1).$$

The point summand has type $(1, 1)$ and removes one middle Tate class. We have proved:

Proposition 3.1. *The generic-twist Hodge vector of IC_X^H is*

$$h(\mathrm{IC}_X^H) = (2, 3, 2).$$

The associated antidiagonal Hodge cocharacter has weights $-2, 0, 2$ on $\omega(\mathrm{IC}_X)$ with multiplicities $2, 3, 2$.

4. THE UNIVERSAL PAIR-INCIDENCE SPLITTING

We now prove the geometric input for pair convolution. All statements in this section are made after restricting to the open loci on which the displayed divisors are reduced and all Abel points have $h^0 = 1$.

4.1. A dominant normalized chart. Let D be the unique reduced degree-three divisor representing a general point $x \in X$. Its norm is a general line section of the plane cubic E . After an affine coordinate change, the line can be written $y = x$. If the three selected lifts have w -coordinates a, b, c , then

$$f(x) - x^2 = \lambda(x - a^3)(x - b^3)(x - c^3), \quad \lambda \neq 0.$$

Put $e_i = e_i(a, b, c)$ and

$$p(W) = W^3 - e_1 W^2 + e_2 W - e_3.$$

Conversely, these formulas reconstruct the curve and the divisor

$$D : \quad x = y = w^3, \quad p(w) = 0.$$

Thus (λ, e_1, e_2, e_3) is a dominant rational chart of the actual space of pairs (C, D) .

There is a scaling redundancy

$$(x, y, w) \mapsto (r^3 x, r^3 y, rw),$$

under which

$$(\lambda, e_1, e_2, e_3) \mapsto (r^{-3} \lambda, r e_1, r^2 e_2, r^3 e_3).$$

On $e_1 \neq 0$ the rational choice $r = e_1^{-1}$ gives the gauge $e_1 = 1$. No root is adjoined, so this normalization does not split a quadratic cover artificially.

4.2. The norm conic. The evaluation map on D has rank three. Hence $H^0(C, 3\kappa - D)$ has dimension three, with basis

$$s_0 = x - y, \quad s_1 = p(w), \quad s_2 = (x - e_3)w - e_1 y + e_2 w^2.$$

For $s = us_0 + vs_1 + zs_2$, write

$$s = \alpha + \beta w + \gamma w^2,$$

where

$$\alpha = u(x - y) + v(y - e_3) - e_1 zy,$$

$$\beta = e_2 v + z(x - e_3),$$

$$\gamma = -e_1 v + e_2 z.$$

Since $w^3 = y$, its cyclic norm is

$$\mathrm{Nm}(s) = \alpha^3 + \beta^3 y + \gamma^3 y^2 - 3\alpha\beta\gamma y.$$

The section (4.2) is divisible on E by the fixed line $y = x$. The quotient belongs to $H^0(E, 6O)$ and is represented by a conic

$$Q_s = q_{00} + q_{01}x + q_{02}y + q_{11}x^2 + q_{12}xy + q_{22}y^2.$$

Let M_s be its symmetric 3×3 matrix.

Proposition 4.1 (Universal norm identity). *Over $\mathbf{Q}(\lambda, e_1, e_2, e_3)$ one has*

$$\det(M_s) = L_{\text{str}} F_8,$$

where $L_{\text{str}} = u - v$ has multiplicity one and F_8 is a homogeneous factor of degree eight. The geometric generic curve $F_8 = 0$ is integral.

On the chart $e_1 = v = 1$, let $F(z)$ be the monic sextic obtained from F_8 and put

$$\Delta = -4(q_{00}q_{11} - (q_{01}/2)^2).$$

There exist polynomials $A, B \in \mathbf{Q}(\lambda, e_2, e_3, u)[z]$ of degrees three and at most two such that

$$F = A^2 + 3\Delta B^2.$$

In particular, the intrinsic double cover ordering the two factors of a rank-two conic splits over the function field of F_8 after base change to $\mathbf{C}$.

Proof. The determinant factorization follows by substituting (4.2), reducing modulo $y^2 = f(x)$, and solving in the basis $1, x, y, x^2, xy, y^2$ of $H^0(E, 6O)$. All operations are over the indicated rational function field.

The residual factor is geometrically integral on a nonempty open. At the specialization

$$(\lambda, e_1, e_2, e_3) = (1, 6, 11, 6), \quad (a, b, c) = (1, 2, 3),$$

its reduction modulo 13 is irreducible of degree eight and has the smooth rational point $(u, v) = (0, 2)$ on the chart used in the certificate. If an irreducible curve over a finite field were geometrically reducible, Frobenius would permute its geometric components transitively; every rational point would lie on every component and would be singular. Thus this fibre is geometrically integral. Openness of geometric integrality in a flat projective family proves the generic assertion.

The exact coefficient formulas and two independent checks of (4.1) are given in Section A. On the cover $t^2 + 3\Delta = 0$, the identity becomes

$$F = (A + tB)(A - tB).$$

The polynomial B is nonzero and has degree at most two, whereas F is irreducible of degree six on this chart; hence B is nonzero in the function field. The two rational roots $t = \pm A/B$ are distinct. Finally, Δ is the discriminant of the restriction of Q_s to a coordinate line. On the rank-two locus, ratios obtained from different nonzero principal minors are squares because $\text{adj}(M_s)$ has rank one. Thus Δ defines the intrinsic ordering character. The constant -3 is a square over $\mathbf{C}$, completing the proof. $\square$

4.3. The scheme-theoretic Abel inverse. We isolate the relative argument that turns ordered conic factors into ordered components of the Abel incidence.

Lemma 4.2 (Relative ordered-zero construction). *Let S be a nonempty open of the unordered distinct rank-two incidence. Let $\mathcal{I}_{\text{fac}} \rightarrow S$ be the finite étale double cover that orders the two conic factors. Let $\mathcal{I}_{\text{div}} \rightarrow S$ be the incidence whose T -points are ordered relative degree-three Cartier divisors (D_1, D_2) with*

$$\mathcal{O}_{C_T}(D + D_1 + D_2) \simeq 3\kappa_T, \quad \pi_* D_i \in |\eta_T^3|,$$

such that the induced projective section of $3\kappa_T$ is the pullback of $[s]$ and the two norm divisors are the residual line sections of Q_s . Impose also the transverse, disjointness, branch-avoidance, and Abel-open conditions used below. After further

shrinking S , the following assignments define inverse S -isomorphisms $\mathcal{I}_{\text{fac}} \simeq \mathcal{I}_{\text{div}}$ and commute with every base change $S' \rightarrow S$, including nonreduced base change:

- (i) factor the residual norm conic as an ordered product $Q_s = \ell_1 \ell_2$;
- (ii) write $\text{div}(s) = D + D_1 + D_2$, where $\text{Nm}(D_i)$ is the line section cut out by ℓ_i .

The divisors D_i are relative effective Cartier divisors of degree three, and, after restricting to the relative smooth Abel open, their Abel classes are smooth points of X .

Proof. For the forward construction put $T = \mathcal{I}_{\text{fac}}$ and write $Q_s = \ell_1 \ell_2$ in its universal ordering. Remove the loci where a line is tangent to E , where either line meets the branch divisor, where any two of the fixed and residual line sections meet, where s vanishes on two sheets over one point of a residual line section, where a norm factor has multiplicity greater than one, or where a relevant Abel class has $h^0 > 1$ or lies outside the relative smooth locus of X/T . The line ℓ_i then cuts out a relative finite étale divisor $B_i \subset E_T$ of degree three. Moreover

$$U_i = C_T \times_{E_T} B_i \longrightarrow B_i$$

is finite étale of degree three, because B_i avoids the branch divisor.

We first select the zero over B_i scheme-theoretically. Étale-locally on B_i , split $U_i = B_i \sqcup B_i \sqcup B_i$ and trivialize the restriction of 3κ , using the induced trivialization of its norm line. If $a_1, a_2, a_3 \in \mathcal{O}_{B_i}$ are the three values of s , compatibility of the finite locally free norm with base change gives

$$a_1 a_2 a_3 = \text{Nm}_\pi(s)|_{B_i} = 0$$

as an equality in the structure sheaf, not merely at geometric points. The collision exclusions imply that locally two values, say a_2, a_3 , are units. Equation (4.3) then forces $a_1 = 0$, also in nilpotent directions. Consequently

$$Z_i = Z(s) \times_{E_T} B_i \subset U_i$$

is locally exactly one open-and-closed sheet of U_i/B_i . Thus $Z_i \rightarrow B_i$ is finite étale of degree one and hence is an isomorphism.

Let t be a local equation for B_i in E_T . Since B_i occurs with multiplicity one in the norm divisor and the other norm factors are disjoint from it, $\text{Nm}_\pi(s) = tu$ with u a unit along B_i . Étale-locally on a neighbourhood of B_i in the complement of the branch divisor, split the cover $C_T \rightarrow E_T$. On the selected sheet the other two values of s remain units, so the same local product formula shows that the selected value is t times a unit. The zero is therefore simple. The image $D_i \subset C_T$ of Z_i is finite étale of degree three over T ; as a finite étale multisection of the smooth relative curve C_T/T , it is a relative effective Cartier divisor.

The pairwise disjoint divisors D, D_1, D_2 lie in $Z(s)$ and have total relative degree nine, equal to $\deg(3\kappa)$. After dividing by their Cartier equations, the residual section has degree zero and is nonvanishing on every geometric fibre. Nakayama's lemma makes it a trivialization, so there is no vertical or nilpotent residual contribution and

$$Z(s) = D + D_1 + D_2$$

as relative Cartier divisors. Conversely, an ordered divisor datum determines the unique projective section $[s]$ with zero divisor $D + D_1 + D_2$, and taking its norm

recovers the ordered line sections B_i and hence the ordered factors ℓ_i . Thus these assignments are inverse already at the divisor level.

For completeness, let

$$\alpha : C_T^{(3)} \longrightarrow \mathrm{Pic}^3(C_T/T)$$

be the relative Abel map and let $F = \mathrm{Nm}^{-1}(\eta^3) \subset \mathrm{Pic}^3(C_T/T)$. The norm fibre is the cartesian base change

$$N_T = C_T^{(3)} \times_{\mathrm{Pic}^3(C_T/T)} F.$$

On the relative Picard scheme, let $W_3^{(1)}$ be the open on which $h^0 = 1$ and the pushforward of a Poincaré bundle is locally free of rank one and commutes with base change; pull this open back to T . The standard projective-bundle description of the Abel map and cohomology and base change then identify its source with the projectivization of that rank-one bundle. Thus the map is an isomorphism and remains so after arbitrary base change. Fibrewise, this is the $h^0 = 1$ case of Riemann–Kempf [11]:

$$\alpha^{-1}(W_3^{(1)}) \xrightarrow{\sim} W_3^{(1)}.$$

Base changing this isomorphism by $F \rightarrow \mathrm{Pic}^3(C_T/T)$ and translating by κ^{-1} gives

$$N_T^{(1)} \xrightarrow{\sim} X_T^{(1)}.$$

We replace both sides by the inverse images of $X_T^{(1)} \cap X_{T,\mathrm{sm}/T}$, as imposed above. Thus the divisor construction descends to the relative smooth Abel incidence.

Zero schemes, fibre products, finite locally free norms, and the cartesian Abel construction all commute with arbitrary base change. Hence both composites preserve the universal zero schemes and are the identity as morphisms. The use of $C_T^{(3)} \simeq \mathrm{Hilb}^3(C_T/T)$ is only representability; no normalization of the ordering cover or of pair fibres is involved. $\square$

The open in [Theorem 4.2](#) is nonempty. After reduction modulo 13, an exact witness is provided by

$$E : y^2 = x^3 - 64x + 64, \quad (a, b, c) = (1, 2, -2), \quad (u : v : z) = (1 : 0 : 9) \pmod{13}.$$

At this point the conic factors over $\mathbf{F}_{13}$ as

$$-(4x + y - 6)(6x + y + 5).$$

The two line-section discriminants, the branch resultants, the pair resultant, and the six evaluations on the fixed divisor are all nonzero; see [Section A](#). The unique g_3^1 is $|\kappa|$. In the coordinates above, the norm of a section of $H^0(C, \kappa) = H^0(E, \eta) \oplus H^0(E, \mathcal{O}_E)$ cuts out a line with no x -term, whereas the three norm lines

$$y - x, \quad 4x + y - 6, \quad 6x + y + 5$$

all have a nonzero x -term. Thus their divisors have $h^0 = 1$ and avoid the contracted pencil. Hence every excluded condition above is a proper closed condition.

4.4. Boundary exhaustion and global labels. Fix a geometric generic pair (C, D_x) . For

$$C_x = \{y \in X : -x - y \in X\},$$

the unique effective divisors representing $x, y, -x - y$ define a rational map

$$\Psi : C_x \dashrightarrow \mathbf{P}H^0(C, 3\kappa - D_x).$$

Its fibres are finite: for a fixed section, the residual zero-cycle has only finitely many degree-three subcycles. Thus every curve component of C_x maps to a component of $\det(M_s) = 0$.

The structural line has the intrinsic description

$$\mathbf{P}H^0(3\kappa - D_x - D_{-x}) = \mathbf{P}H^0(\kappa).$$

Its members have divisor $D_x + D_{-x} + D_0$ with $D_0 \in |\kappa|$. For two reduced disjoint line sections of a plane cubic, a mixed degree-three subcycle is not a line section: the line through two points of either triple contains the third point of that same triple. Hence the structural line represents only the fixed pair $\{-x, 0\}$ and does not give a curve component. The indeterminacy locus $y = 0$ or $-x - y = 0$ is also zero-dimensional. Since F_8 is the only remaining determinant component, [Theorem 4.2](#) exhausts all nonexceptional curve components. We conclude that the geometric generic C_x is reduced and has exactly two integral components A_x, B_x , exchanged by

$$y \mapsto -x - y.$$

Reducedness follows because C_x is a local complete intersection curve in the smooth threefold P , hence Cohen–Macaulay, and is reduced at both generic points.

It remains to descend the two labels from the normalized chart. Let τ be the order-three deck transformation. On P one has

$$1 + \tau + \tau^2 = 0,$$

and $\tau(X) = X$. Therefore

$$s_+(x) = (\tau x, \tau^2 x) \in C_x$$

is a rational section of the pair-incidence family. It is smooth for general x . Otherwise the Gauss covectors at τx and $\tau^2 x$ would agree; equivariance would force the general Gauss image into the projective fixed locus of τ , contrary to [\(3.2\)](#). Explicitly,

$$T_0^*P = \left\langle \frac{dx}{w^4} \right\rangle \oplus \left\langle \frac{dx}{w^5}, \frac{x dx}{w^5} \right\rangle,$$

and the two summands have distinct τ -eigenvalues. The projective fixed locus is a point union a line, a proper subset of $\mathbf{P}^2$.

Over the function field of the actual pair space, a smooth rational point lies on exactly one geometric component. Galois therefore fixes the component containing s_+ and, since there are exactly two components, also fixes the other. Taking closures and using generic flatness and openness of geometric integrality and reducedness gives the following.

Theorem 4.3 (Pair splitting). *After shrinking to a nonempty open subfamily of curves, there is a dense connected open $U \subset X_{\text{reg}} \setminus \{0\}$ such that, for every $x \in U$, the nonexceptional part of C_x has exactly two reduced integral curve components A_x, B_x . They are fibres of two globally labelled flat relative components over U , and factor transposition exchanges them.*

5. PAIR CONVOLUTION DETECTS IC_X

Let $\rho : S \rightarrow X$ be the minimal resolution and let $j : S \rightarrow P$ be its composition with the inclusion. Put

$$K = \mathrm{IC}_X, \quad L = Rj_* \mathbf{Q}_S[2] = K \oplus \delta_0.$$

Consider

$$\tilde{m} : S \times S \longrightarrow P, \quad (p, q) \longmapsto j(p) + j(q),$$

and let $\sigma(p, q) = (q, p)$. We use the minus sign to denote the anti-invariant direct summand under σ .

Proposition 5.1 (Alternating support detector). *Under the conclusion of Theorem 4.3, one has a direct summand*

$$K \subset {}^p H^0(Rm_*(K \boxtimes K))_-.$$

Consequently K is a direct summand of $\mathrm{Alt}^2(K)$ in the convolution Tannakian category.

Proof. Put $U^+ = -U$. Fix $t \in U^+$, write $x = -t \in U$, and let $E_{\mathrm{exc}} = \rho^{-1}(0)$. By definition, C_x parametrizes ordered pairs whose sum is $-x = t$. Hence the fibre $F_t = \tilde{m}^{-1}(t)$ has precisely four irreducible curve components: the strict transforms of A_x, B_x and the two structural components

$$C_{1,t} = E_{\mathrm{exc}} \times \{s_t\}, \quad C_{2,t} = \{s_t\} \times E_{\mathrm{exc}},$$

where s_t is the unique point above t . Indeed, away from E_{exc} the resolution is an isomorphism, while a pair with one entry on E_{exc} must have the other entry over t . Thus there are no further top-dimensional components. The involution σ exchanges A_x with B_x and $C_{1,t}$ with $C_{2,t}$.

Proper base change identifies

$$\mathcal{H}^{-2}(L * L)_t = H^2(F_t, \mathbf{Q}).$$

The top cohomology of a proper complex curve is freely generated by the fundamental classes of the irreducible components of its reduction. Holomorphic transposition preserves complex orientation. The two exchanged pairs therefore give

$$\dim \mathcal{H}^{-2}(L * L)_{-,t} = 2.$$

Expand the decomposition $L = K \oplus \delta_0$:

$$L * L = (K * K) \oplus (K * \delta_0) \oplus (\delta_0 * K) \oplus (\delta_0 * \delta_0).$$

At $t \neq 0$, the last term has zero stalk. The two middle terms are both canonically K , and transposition exchanges them. Since t is a smooth point of the surface X , their anti-invariant stalk in degree -2 has dimension one. Subtracting this from (5) gives

$$\dim \mathcal{H}^{-2}(K * K)_{-,t} = 1.$$

There is no Koszul sign: K has geometric shift $[2]$, and $(-1)^{2 \cdot 2} = 1$.

After a topological stratification and a further shrinking of U^+ , the four globally labelled relative components trivialize the full alternating top cohomology local system. The anti-diagonal of the two cross terms is also a constant local subsystem. Hence the rank-one local system in (5) is constant.

For general $t \in P \setminus X$, the fibre of pair addition is the smooth connected curve $X \cap (t - X)$. It is connected because it is the intersection of two ample divisors in

the smooth projective threefold P . The involution $p \mapsto t - p$ preserves its complex orientation and hence acts by $+1$ on its one-dimensional top cohomology. Therefore

$$\mathcal{H}^{-2}(K * K)_- = 0 \quad \text{on a dense open subset of } P.$$

The complex $(K * K)_-$ is a direct summand of the projective direct image of a pure polarizable Hodge module. By the decomposition theorem and semisimplicity [1, 19], it is a direct sum of shifted simple intersection complexes. A summand with proper support containing the generic point of U^+ must have support X . Since $\mathrm{IC}_X(\mathcal{V})|_{U^+} = \mathcal{V}[2]$, contribution to ordinary degree -2 forces perverse shift zero.

We spell out why a full-support summand cannot account for the line in (5). At a general point of U^+ choose a normal analytic disk Δ to X , and write $j_\Delta : \Delta^* \hookrightarrow \Delta$. Locally, a simple perverse intersection complex with support P has the form

$$\mathcal{L}[2] \boxtimes (j_{\Delta,!} \mathcal{W}[1])$$

for local systems $\mathcal{L}$ and $\mathcal{W}$. Writing $i_0 : \{0\} \hookrightarrow \Delta$, the defining no-quotient property of intermediate extension gives

$$\mathcal{H}^0(i_0^* j_{\Delta,!} \mathcal{W}[1]) = 0.$$

Since a perverse sheaf on a disk has ordinary stalk amplitude $[-1, 0]$, $i_0^* j_{\Delta,!} \mathcal{W}[1]$ is concentrated in degree -1 when nonzero. Thus an unshifted full-support summand can contribute there only in ordinary degree -3 . Contribution in degree -2 requires the perverse-degree-one shift $[-1]$. But on the dense smooth open of P that same summand is a nonzero local system shifted by $[3](-1)$ and therefore contributes in degree -2 , contrary to (5). Supports of dimension at most one do not meet a generic point of U^+ , and a different irreducible divisor cannot contain U^+ .

Thus the constant rank-one local system in (5) is the coefficient system of an unshifted IC_X summand. This proves the first assertion. The even support dimension identifies the anti-invariant convolution summand with the ordinary exterior square under the fibre functor. Finally, generic vanishing makes the nonzero perverse-cohomological degrees of convolution negligible; the semisimple quotient is exact and preserves direct summands. Since $\chi(K) = 7$, the displayed K remains nonzero in the quotient. This proves the second assertion. $\square$

6. IDENTIFICATION OF THE FULL TANNAKA GROUP

Let

$$G = G(K), \quad V = \omega(K).$$

By (3.2), $\dim V = 7$. The symmetry (3.1), Verdier self-duality, and the even support dimension give a perfect symmetric evaluation pairing

$$K * K \longrightarrow \delta_0.$$

It takes values in the actual tensor unit because the centre of the symmetry is zero. Therefore the full group embeds faithfully as

$$G \hookrightarrow O(V).$$

The simple perverse sheaf K is nondivisible: a translation fixing it must fix its support, and (3.2) then makes the translation zero. By [9, Proposition 3.3],

$$V|_{G^0} \quad \text{is irreducible.}$$

This result does not require the ambient abelian variety to be simple.

Let $G_P = G$ and let G_H be the Tannaka group of the pure Hodge module IC_X^H , acting faithfully on the same complex vector space V . By [12, Corollary 3.2], the Hodge decomposition is defined by a cocharacter $\lambda : \mathbf{G}_m^2 \rightarrow G_H$. The comparison theorem [12, Corollary 2.5] gives

$$G_P \triangleleft G_H, \quad [G_P^0, G_P^0] = [G_H^0, G_H^0] =: G^*.$$

Since $G_P^0 \subset G_H^0$, equation (6) implies that $V|_{G_H^0}$ is irreducible. Thus the connected centre Z_H^0 acts by scalar matrices. As G_H^0 is reductive, multiplication $Z_H^0 \times G^* \rightarrow G_H^0$ is surjective with finite kernel.

Put $\rho(z) = \lambda(z, z^{-1})$. The source of λ is connected, so $\rho(\mathbf{G}_m) \subset G_H^0$. By Theorem 3.1, $\det \rho = 1$. The determinant is trivial on the semisimple group G^* and therefore descends to a character

$$\overline{\det} : G_H^0/G^* \rightarrow \mathbf{G}_m.$$

The quotient G_H^0/G^* is the image of Z_H^0 . On Z_H^0 the determinant is $c \mapsto c^7$; since the faithful scalar action embeds Z_H^0 into $\mathbf{G}_m$, the induced character $\overline{\det}$ has finite kernel. Hence the image of the connected group $\mathbf{G}_m$ under ρ in G_H^0/G^* is both connected and finite, and is therefore trivial. Consequently

$$\rho(\mathbf{G}_m) \subset G^*,$$

the common Hodge and perverse derived group. Its weights on V are

$$-2, 0, 2 \quad \text{with multiplicities } 2, 3, 2.$$

Applying the fibre functor to Theorem 5.1 gives

$$\mathrm{Hom}_G(V, \Lambda^2 V) \neq 0.$$

Proposition 6.1. *The identity component of G is G_2 in its standard representation.*

Proof. By (6) and (6), the connected centre of G^0 acts by orthogonal scalars and is therefore trivial. Thus G^0 is semisimple. An irreducible representation of a product of semisimple groups is an external tensor product. Since 7 is prime and the action is faithful, G^0 has one simple factor.

The classification of irreducible self-dual seven-dimensional representations leaves exactly

$$\begin{aligned} &SO_7 \text{ on its standard module,} \quad G_2 \text{ on its standard module,} \\ &PGL_2 \text{ on } \mathrm{Sym}^6(\mathbf{C}^2). \end{aligned}$$

Passing to the simply connected cover and consulting the low-dimensional highest weights gives this list directly [5, 7]: type A_6 also has a seven-dimensional standard module, but it is not self-dual, and the remaining simple types have no irreducible module of dimension seven. For SO_7 , the exterior square of the standard module is the irreducible adjoint module, so $\mathrm{Hom}_{SO_7}(V, \Lambda^2 V) = 0$, contrary to (6). For a nontrivial cocharacter of PGL_2 , the weights on $\mathrm{Sym}^6(\mathbf{C}^2)$ are seven distinct numbers. For a primitive cocharacter they are

$$-3, -2, -1, 0, 1, 2, 3,$$

and every other nonzero cocharacter multiplies them by a nonzero integer. The trivial cocharacter has one weight of multiplicity seven. Neither possibility agrees with (6). Hence $G^0 = G_2$. $\square$

It remains to identify the component group. In the standard orthogonal representation,

$$N_{O_7}(G_2) = G_2 \times \{\pm I\}.$$

Indeed, G_2 has no outer automorphisms, and its centralizer on the irreducible standard module consists of scalars. If G met the second coset, it would contain $-I$. But $-I$ acts by -1 on V and by $+1$ on $\Lambda^2 V$, forcing every morphism in (6) to vanish. This contradiction proves $G = G_2$.

7. THE NONEMPTY OPEN FAMILY

We now make the quantifier in [Theorem 1.1](#) explicit. Over the regular integral base $\mathcal{B}$, let $\mathcal{C} \rightarrow \mathcal{B}$ be the relative family of cyclic covers, let $\mathcal{P} \rightarrow \mathcal{B}$ be the relative Prym abelian scheme, and let $\mathcal{X} \subset \mathcal{P}$ be the relative centred theta section. The divisor $\mathcal{X}$ is flat over $\mathcal{B}$: it is a relative effective Cartier divisor in the smooth scheme $\mathcal{P}$, and its defining section remains a non-zero-divisor on every fibre. The preceding geometry applies after arbitrary geometric base change: every fibre is geometrically integral and its only singular point is the zero section. Put

$$\mathcal{M} = (\mathcal{X}/\mathcal{B})_{\text{sm}} = \mathcal{X} \setminus 0.$$

Then $\mathcal{M} \rightarrow \mathcal{B}$ is smooth and surjective, and every fibre is a dense geometrically integral open of the corresponding theta surface. Moreover, $\mathcal{M}$ is integral. Indeed, since $\mathcal{B}$ is regular and $\mathcal{M} \rightarrow \mathcal{B}$ is smooth, $\mathcal{M}$ is regular, so its irreducible components are open. The image of each component is a nonempty open of the integral base $\mathcal{B}$ and hence contains its generic point. Each component would therefore yield an irreducible component of the geometrically integral generic fibre. Thus there is only one.

Let $\mathcal{N}$ be the relative norm fibre in $\mathcal{C}^{(3)}$. On $\mathcal{M}$ every Abel class has $h^0 = 1$, so the relative Abel isomorphism used in [Theorem 4.2](#) identifies $\mathcal{N} \times_{\mathcal{X}} \mathcal{M}$ with $\mathcal{M}$. The universal divisor on the relative symmetric cube therefore supplies the universal effective divisor over $\mathcal{M}$. The unnormalized four-parameter chart (λ, e_1, e_2, e_3) of [Section 4](#), before imposing the gauge $e_1 = 1$, maps dominantly to $\mathcal{M}$. After restricting its domain, let $T \rightarrow \mathcal{M}$ be an integral dominant chart. Equivalently, one may reintroduce the scaling parameter $r \in \mathbf{G}_m$ into the slice $e_1 = 1$; that slice is used only for the function-field calculation and does not lower the dimension of the dominant chart. After the standard local-freeness shrinking, write $p : \mathcal{C}_T \rightarrow T$ for the relative curve and D for its universal degree-three divisor. Then $p_* \mathcal{O}_{\mathcal{C}_T}(3\kappa - D)$ is a rank-three vector bundle.

All charts, the residual equation, and the excluded-locus functions are defined over an integral localization

$$R_0 = \mathbf{Z}[1/N],$$

where $13 \nmid N$ and every displayed denominator is invertible at the two modulo-13 witnesses. Denote the resulting integral model of T and its relative curve by T_{R_0} and $p_{R_0} : \mathcal{C}_{T_{R_0}} \rightarrow T_{R_0}$. After shrinking around the relevant charts, the structure map $T_{R_0} \rightarrow \text{Spec } R_0$ is dominant. Since R_0 is a principal ideal domain and T_{R_0} is integral, its affine coordinate rings are R_0 -torsion-free; hence T_{R_0} is R_0 -flat. The residual degree-eight Cartier divisor

$$\mathcal{R}_{R_0} \subset \mathbf{P}_{T_{R_0}}(p_{R_0*} \mathcal{O}_{\mathcal{C}_{T_{R_0}}}(3\kappa - D))$$

is projective and flat over T_{R_0} . The first modulo-13 certificate in [Theorem 4.1](#) is a geometrically integral fibre of this family. Geometric integrality is open for a flat projective family; the corresponding nonempty open of the integral scheme T_{R_0} contains its generic point. After base change to $\mathbf{C}$ and a further shrinking of T , it follows that $\mathcal{R} = \mathcal{R}_{R_0} \times_{R_0} \mathbf{C}$ is integral with geometrically integral fibres.

On this single integral space $\mathcal{R}$, reducedness of the fixed and residual degree-three divisors, rank exactly two of the norm conic, distinctness of its factors, finite étaleness and disjointness of the line sections, and avoidance of the branch divisor and contracted pencil are open conditions. Let g be the product of the corresponding discriminants, resultants, and evaluations on the affine witness chart. The model $\mathcal{R}_{R_0}$ is flat over R_0 by composition, so its affine coordinate rings are R_0 -torsion-free. The second certificate gives $g \neq 0$ at an $\mathbf{F}_{13}$ -point of $\mathcal{R}_{R_0}$; hence g is nonzero on the characteristic-zero generic fibre. Consequently the good locus $\mathcal{R}_g \subset \mathcal{R}$ is a nonempty open. The universal identity (4.1) splits the ordering cover at the generic point of $\mathcal{R}$, and the relative ordered-zero construction identifies its ordered sheets with the two Abel components. Let $\mathcal{I} \rightarrow \mathcal{M}$ denote the relative pair-incidence family, whose fibre over x is C_x . Since $T \rightarrow \mathcal{M}$ is dominant, the preceding calculation after base change to T shows that the geometric generic fibre of $\mathcal{I} \rightarrow \mathcal{M}$ has exactly two geometrically integral reduced components. The $\mathbf{C}(\mathcal{M})$ -rational section $x \mapsto (\tau x, \tau^2 x)$ is a smooth point of that generic fibre and hence lies on exactly one geometric component. Galois fixes this component and, since there are exactly two, also fixes the other. Thus both components descend to the function field of $\mathcal{M}$. Take their closures $\mathcal{A}_1, \mathcal{A}_2 \subset \mathcal{I}$. By generic flatness and openness of geometric integrality and reducedness, after replacing $\mathcal{M}$ by a nonempty open $\mathcal{M}^\circ$, the two closures are flat with geometrically integral reduced curve fibres, transposition exchanges them, and their union is the full nonexceptional pair fibre. Shrinking also removes every vertical component.

The morphism $\mathcal{M}^\circ \rightarrow \mathcal{B}$ is the restriction of a smooth morphism and is therefore open. Its image

$$\mathcal{B}^\circ = \text{im}(\mathcal{M}^\circ \rightarrow \mathcal{B})$$

is consequently a nonempty Zariski-open subset. For every $b \in \mathcal{B}^\circ(\mathbf{C})$, the fibre $U_b = \mathcal{M}^\circ \cap X_b$ is a nonempty open of the integral surface X_b ; hence it is dense and connected. The pair-splitting theorem holds over U_b , and all arguments of the preceding sections apply. This proves [Theorem 1.1](#) with the stated “for every b ” quantifier.

APPENDIX A. EXACT NORM CERTIFICATE

We give enough information to reproduce the only large symbolic identity in the proof. On $e_1 = v = 1$, write

$$F = z^6 + c_5 z^5 + c_4 z^4 + c_3 z^3 + c_2 z^2 + c_1 z + c_0$$

and put $k = 3\Delta$. Let L be the leading numerator of the unnormalized sextic obtained from the determinant calculation. Set

$$A = z^3 + a_2 z^2 + a_1 z + a_0, \quad B = b_2 z^2 + b_1 z + b_0,$$

where

$$\begin{aligned} b_2 &= \frac{e_2\lambda}{2L}(2e_2^3\lambda u + e_2^3\lambda - 6e_2e_3\lambda u + 6e_2e_3\lambda - 3e_3\lambda + 3e_3), \\ b_1 &= -\frac{\lambda}{2L}(e_2^3\lambda u + 2e_2^3\lambda + 3e_2^2e_3\lambda u^2 - 6e_2^2e_3\lambda u + 3e_2^2e_3\lambda \\ &\quad - 3e_2e_3\lambda u + 3e_2e_3\lambda - 3e_3\lambda + 3e_3), \\ b_0 &= \frac{\lambda(u-1)}{2L}(-e_2^3\lambda + 3e_2e_3\lambda u - 3e_2e_3\lambda + 2e_3\lambda - 2e_3), \end{aligned}$$

and

$$a_2 = \frac{c_5}{2}, \quad a_1 = \frac{c_4 - a_2^2 - kb_2^2}{2}, \quad a_0 = \frac{c_3}{2} - a_2a_1 - kb_2b_1.$$

Substitution proves $F = A^2 + kB^2$. The verifier performs this substitution in the exact fraction field $\mathbf{Q}(\lambda, e_2, e_3, u)$ and obtains seven zero coefficient remainders. An independent verifier clears the common denominator $4\lambda^2L^2$ and obtains the zero polynomial, with output

```
A_NUMERATOR_TERMS      117,
B_NUMERATOR_TERMS      24,
CURVE_NUMERATOR_TERMS  248,
FOURS_NUMERATOR_TERMS  20,
DIRECT_NUMERATOR_TERMS  0.
```

No random specialization or floating-point operation enters either identity check.

The geometric-integrality certificate at $(a, b, c) = (1, 2, 3)$, modulo 13, reports one nonconstant factor of degree eight and exponent one, together with

$$\text{POINT_VALUE} = 0, \quad \text{POINT_GRADIENT} = (-2, 0).$$

The genuine-open certificate for the second sample reports

```
FIXED_POINT_LINE_VALUES = (-1, 34, 19, 12, 61, 40),
LINE_SECTION_DERIVATIVE_RESULTANTS = (-3, 6),
LINE_BRANCH_RESULTANTS = (-1, 1),
LINE_PAIR_RESULTANT = -1,
```

all in $\mathbf{F}_{13}$ and hence all nonzero. The source equations, coefficient formulas, and verification scripts are the supplementary files `code/g2_mu6_source_provenance_verify.py`, `code/g2_mu6_lowmem_certificate_fast.py`, `code/g2_mu6_direct_numerator_verify.py`, and their inputs in `code/generated/`. The provenance verifier derives the saved Singular inputs from the normalized elliptic cubic and the three sections used above, then compares the resulting formulas and complete files. The scripts use exact SymPy arithmetic and can be run with

```
python code/g2_mu6_source_provenance_verify.py
python code/g2_mu6_lowmem_certificate_fast.py
python code/g2_mu6_direct_numerator_verify.py
```

With Singular installed in WSL, the finite-field factorization and smooth-point checks are reproduced by

```
$audit = "code/g2_mu6_universal_special123_mod13_audit.ps1"
powershell -NoProfile -ExecutionPolicy Bypass -File $audit
```

The complete source and versioned release record are available at <https://github.com/FDmd233/litt-problem-13-g2>.

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